AMD DEVELOPER HACKATHON: ACT II · TRACK 1

Token-Efficient General-Purpose AI Agent

One agent. Eight task categories. The fewest tokens on the leaderboard.

Team ZaynDev · Nairobi, Kenya · github.com/paulandrew12/zayndev-amd-act2-agent

THE CHALLENGE

One agent, eight jobs, judged on tokens

Track 1 asks for a single agent that handles **eight capability categories** — factual Q&A, math, sentiment, summarisation, NER, code debugging, logic, and code generation — using Fireworks-hosted models, shipped as a Docker container.

Pass a binary gate, then win a token race — an objective, leaderboard-scored race.

1 · Accuracy gate

An LLM-judge scores every answer. Miss the threshold and you're off the leaderboard entirely.

2 · Fewest tokens win

Everyone who clears the gate is then ranked by total tokens. Lowest count takes the prize.

THE INSIGHT

Token count is dominated by output verbosity — not by which model you pick.

Left alone, these models “think out loud” before answering. That preamble does two kinds of damage:

Wastes tokens

Hundreds of tokens of reasoning you're billed for on every single task.

Fails the gate

The token cap truncates the reply before the actual answer arrives — an automatic miss.

```
classify sentiment →

"The user wants me to classify the
sentiment. Let me analyze each
clause for emotional valence…"

✗ truncated — no label — FAIL
```

ARCHITECTURE

A three-stage pipeline

01

Zero-token router

Pure-regex classifier sorts each task into one of 8 categories. No LLM call — costs 0 tokens.

0 tokens

→

02

Model per category

Reads ALLOWED_MODELS at runtime and picks the strongest allowed model for that category.

best fit

→

03

Answer-only prompt

A terse, category-specific prompt forbids preamble; a tight cap backstops truncation.

fewest tokens

Container contract: reads /input/tasks.json → writes /output/results.json · runs tasks concurrently within the 10-minute budget

WHY IT'S EFFICIENT

Three levers, one goal: fewer tokens

Zero-token routing

Classification is pure regex. Competitors who route with an LLM call pay for it twice — the routing call plus the answer.

Preamble suppression

Answer-only prompts cut the chain-of-thought that inflates every response — the single biggest lever on the score.

Strongest model, per task

Accuracy headroom is free under the gate. We spend tokens on the answer, never on a model hedging or second-guessing.

VALIDATED END-TO-END, LIVE

The pipeline runs — and we can measure it

98%

router accuracy
(0 tokens spent)

95%

task accuracy, live
on public models

0

tokens spent
routing each task

48MB

image · <60s start
(limit 10GB)

Full 40-task eval, real Fireworks API, real token counts. *Final leaderboard tokens are tuned on the hackathon's own models.*

THE EDGE

Most entries optimise the wrong thing

The common approach

- Fine-tune a router to pick a cheap vs. expensive model
- Send the raw task prompt straight to the model
- Let the model answer however verbosely it likes
- Result: output length quietly dominates the token bill

Ours

- Route for free — regex, zero tokens, no second call
- Per-category answer-only prompts kill the preamble
- Attack the real cost: how long the answer is
- Strongest allowed model — accuracy headroom is free

A fancy router can tie “always use the cheap model.” The untapped lever is verbosity.

ENGINEERING

Built to the spec, built not to break

linux/amd64 manifest

Cross-built for the judging VM — the silent-failure requirement, handled.

48MB, <60s cold start

Tiny image, well under the 10GB cap; ready long before the 60-second limit.

Contract-perfect I/O

Always writes valid /output/results.json — malformed output scores zero.

Fails gracefully

Per-task isolation + retries: one bad task can never sink the whole run.

Async concurrency

Tasks run in parallel, comfortably inside the 10-minute runtime budget.

MIT & reproducible

Public repo, clean modules, one-command local test harness.

DISCIPLINE

We measure before we spend a submission

A scoring harness that mirrors the judge

- 40 labelled tasks spanning all 8 categories
- Checkers mirror the accuracy gate — including sandboxed execution of generated code
- Offline mode (routing + token estimate) and live mode (real accuracy + real tokens)
- Per-category accuracy and token report on every run

10

submissions per hour

So we don't waste them guessing. Every submission is a measured step, not a shot in the dark.

Pass the gate.

Spend the fewest tokens.

A general-purpose agent engineered for one number: total tokens.

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